the atticus project

AI learning how to read like humans

What is NLP?

From previous lessons, you might have learned that NLP is AI that is trained to understand natural language

NLP will allow the AI to break down human text into something the AI can understand

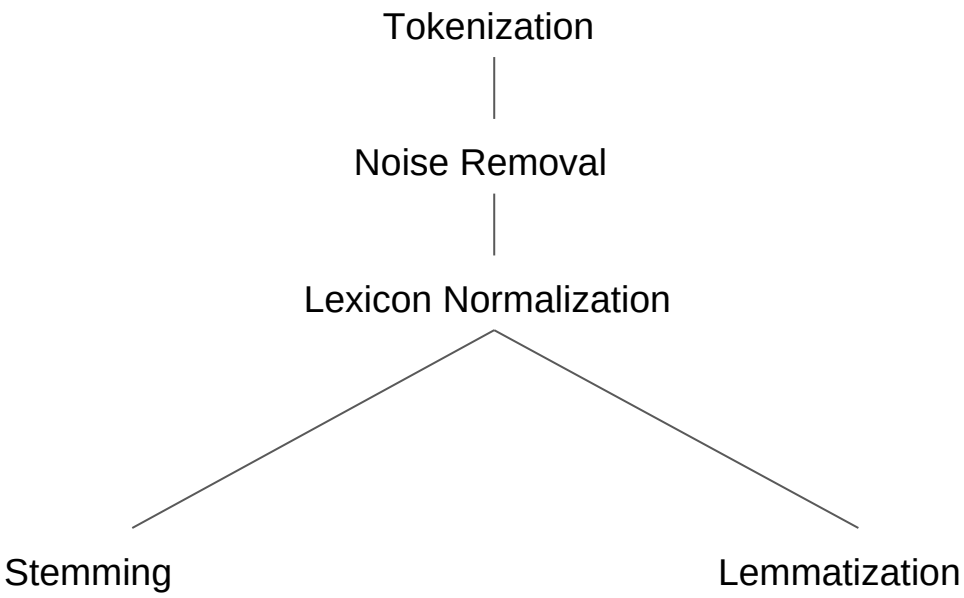
NLP Processing

Text Preprocessing

Feature Engineering

Text Classification

Text Preprocessing



Tokenization

Sentence tokenization:
Breaks down a paragraph into sentence

Word tokenization:
Breaks down a sentence into words

Watch how this paragraph gets decomposed.

Noise Removal

Remove Stop Words

Words that have no meaning and only used for grammatical syntax

“The”, “a”, “an”, etc. You can choose your stop words list

Punctuation can also be removed.

Lexicon Normalization

Stemming

Strip the suffixes such as ing, ly, es, s, etc from a word

It’s rule-based process.

Lemmatization

This process changes words to their base form

It makes use of vocabulary (dictionary importance of words) and morphological analysis (word structure and grammar relations)

Feature Engineering

Features need be generated to analyze the pro-processed data

Entity Extraction - Grouping words into predefined categories.

Syntactical Parsing - Analyzing words and conforming them to normal grammar.

Statistical Features - Keep track of how much words show up.

Entity Extraction

Entity Extraction

Words are grouped based on predefined categories

N-Grams as Features

N is a numeric variable

A combination of N words together

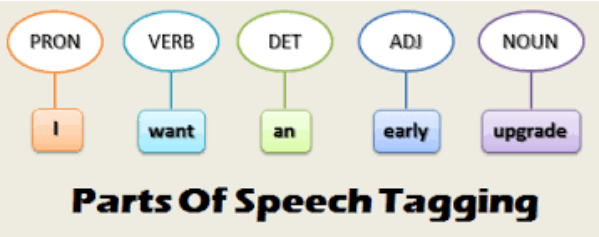
Bigrams (N=2) are considered to be the most important

Syntactical Parsing

Part of Speech Tagging

Every word gets a tag corresponding to its part of speech

Helps AI group words together



Statistical Features

Bag of Words

Keeps track of the number of times a word shows up in a sentence

TF-IDF (Term Frequency - Inverse Document Frequency)

Similar to Bag of words

Uses a formula to measure relevance

Higher TF - IDF number = more relevance to document.

More unique words.

Note: This is log base 2

TFIDF

For a term i in document j:

$$w_{i,j} = tf_{i,j} \times \log\left(\frac{N}{df_i}\right)$$

tf_{ij} = number of occurrences of i in j
df_i = number of documents containing i
N = total number of documents

Bag of Words example

Clause 1: Laws State New York govern matter arise relate Agreement.

Clause 2: Agreement governed internal law, not law regard conflict law, State North Carolina

Clause 3: Agreement governed construed accordance with laws State California without giving effect conflict laws principles

TF-IDF

Clause 1: Laws State New York govern matter arise relate Agreement.

Clause 2: Agreement governed internal law, not law regard conflict law, State North Carolina

Clause 3: Agreement governed construed accordance with laws State California without giving effect conflict laws principles

0.477 is a high number. In this case, the word “matter” is more relevant and unique to Clause 1.

Identification

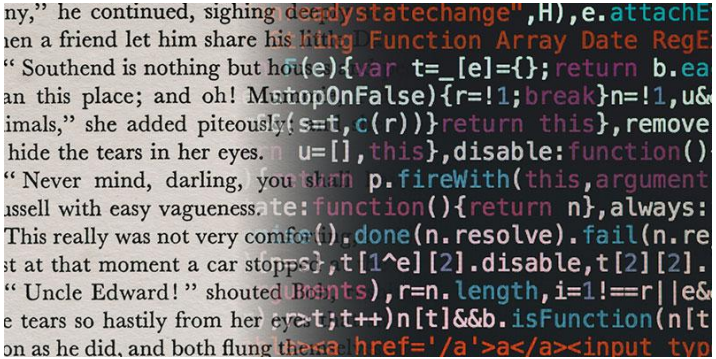
Given that Clause 1 and 2 are governing law clauses, the AI would think that:

“California”, “North Carolina”, and “New York” are all locations (Entity Extraction)

The first four words have a low TF-IDF score, meaning they are shared across many documents. This means the words appear in documents like those(TF-IDF)

There is usually a word which is a location, and it is usually the answer.

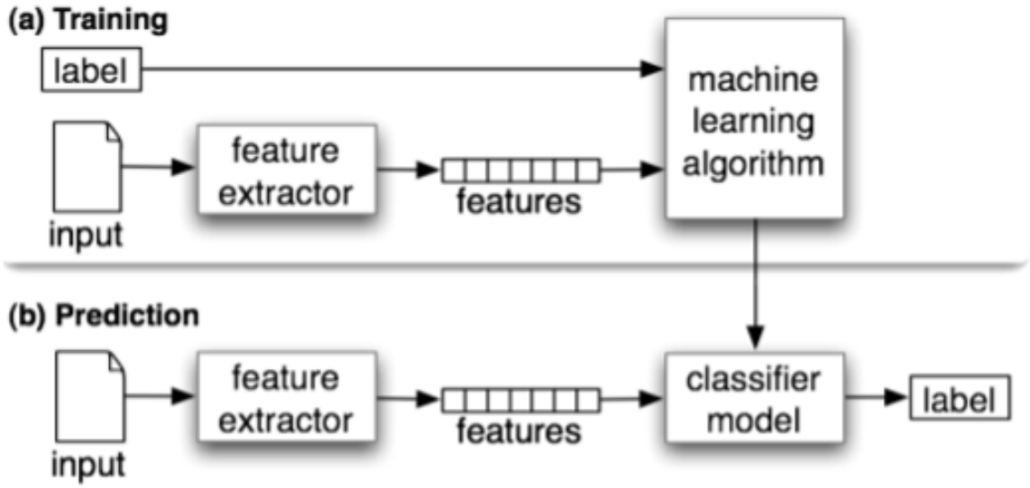
So, the AI would say that Clause 3 is a governing law clause, and that the answer is “California”



Text Classification

A technique to systematically classify a text object such as document or sentence in one of the fixed categories.

A typical natural language classifier consists of two parts: training and prediction. First the text input is processed and features are created. The machine learning models learn the features and is used for prediction against the new text.



Text Classification

```
training_corpus = [
(“This Development Agreement is entered as of this 13th Day of September 2019”, 'Effective Date’),
("Agreement made this 9th Day of March 2020", ‘Effective Date'),
(“This Agreement shall be in effect until March 13, 2021”, 'Expiration Date’),
(“The Term of this Agreement shall commence on the Commencement Date and shall terminate on June 4 2020”, 'Expiration Date')]
test_corpus = [
(“This term of this Agreement shall continue for one (1) year following the Launch Date"), --> 'Expiration Date'
("This Sponsored Research and and License Agreement is entered into on November 28, 2018"), --> 'Effective Date’
("The term of this Agreement shall commence on the Effective Date and shall expire on the last day of the Royalty Term"), --> 'Expiration Date'
```